

Dang Pham (Phạm Cao Đăng)

Astrophysical & Planetary Science and JILA
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ACADEMIC EMPLOYMENT

University of Colorado Boulder, McCray Prize Postdoctoral Fellow	2025 - Present
Cornell University, Research Associate	Sep-Dec, 2020

EDUCATION

University of Toronto	2020 - 2025
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PhD in Astronomy and Astrophysics

- *Advisor:* Professor Hanno Rein
- *Thesis:* Small Bodies, Small Stars: The Fate of Comets
- Thesis awarded the IAU Thesis Prize Honourable Mention (Fundamental Astronomy)

Cornell University , College of Arts and Sciences	2016 - 2020
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Bachelor of Arts in Physics, Mathematics

- *Thesis Advisor:* Professor David Chernoff
- *Senior Thesis:* Bounded cosmic string backreaction and connecting Green's functions for the wave equation

RESEARCH INTERESTS

- *White dwarfs:* pollution, binaries, accretion, magnetism
- *Exoplanet and solar systems:* dynamics, post-main-sequence systems, chemical evolution, formation, characterization and discovery
- *Computational methods:* N-body, integrators, high-performance computing optimization
- *Astrostatistics:* hierarchical Bayesian inference, sampling methods, model selection

AWARDS & HONORS

IAU Thesis Prize Honourable Mention (Division A: Fundamental Astronomy)	2026
Richard McCray Prize Postdoctoral Fellowship	2025 - Present
Ontario Graduate Scholarship	2024
University of St. Michael's College (University of Toronto) Junior Fellowship	2023 - 2025
International Student Fellowship for Excellence in Doctoral Studies	2022
Frank S. Hogg Memorial Fellowship	2021
Summer Research Fellowship Max-Planck-Institut für Astronomie (MPIA)	2019

SEMINARS & CONFERENCES

AT INSTITUTIONS

1. Trotter Space Institute Seminar, McGill University	2026
2. TAPIR Seminar, California Institute of Technology	2026
3. Exoplanet Seminar, University of Wisconsin-Madison	2026
4. Stellar & Exoplanet Seminar, Yale University	2026
5. Seminar, Southwest Research Institute (SwRI)	2026
6. Seminar, Michigan State University	2026
7. Planetary and Star Formation Seminar, University of Michigan	2026
8. Lunch Seminar, University of Colorado Boulder	2025
9. Seminar, University of California Los Angeles	2024
10. Seminar, Harvey Mudd College	2024
11. Planetary Science Seminar, California Institute of Technology	2024
12. Astrophysics Coffee Talk, Institute of Advanced Study	2024
13. Exoplanet Seminar, Princeton University	2024
14. Lunch talk, Harvard Institute for Theory and Computation	2024
15. Seminar, Massachusetts Institute of Technology	2024
16. Lunch Seminar, University of Toronto	2024, 2025
17. Seminar, Institute of Science and Technology Austria	2024
18. Statistics and Machine Learning Seminar, University of Toronto	2022

AT CONFERENCES

19. Multiple Planets Systems Conference, Sofia, Bulgaria.	2026
20. 24th European White Dwarf Conference, Vienna, Austria.	2026
21. REBOUND Conference, Virtual	2024
22. 55th Meeting of the Division of Dynamical Astronomy, Toronto, Canada.	2024
23. 23rd European White Dwarf Conference, Barcelona, Spain.	2024
24. Emerging Researchers in Exoplanet Science VII, Pennsylvania, USA.	2022
25. Great Lakes Exoplanet Area Meeting, Ohio, USA.	2022

COMPUTING ALLOCATIONS AWARDED

Long-term stability of the solar system (PI, ACCESS #PHY260085)	2026 -
• 1.6 million CPU-core hours on TAMU ACES.	
Constraining white dwarf kicks from binary population synthesis (PI)	2026 - 2027
• 450,000 CPU-core hours on Alpine at CU Boulder.	

STUDENT SUPERVISION

UNDERGRADUATE STUDENTS

- Hayden Monk (Michigan State University) 2026 - Present
- *Project*: Efficiency of interstellar objects generation.
- Skyler McCabe and Blake Carver-Adames (CU Boulder) 2026 - Present
- *Project*: Implementing gravitational structures in REBOUND.
 - Project is awarded the Astrophysical & Planetary Science Undergraduate Research Fellowship (\$3000).
- Thomas Alexander (CU Boulder) 2025 - Present
- *Project*: Oort clouds after white dwarf natal kicks.
 - Project is awarded the Undergraduate Research Opportunities Program (UROP) grant by CU Boulder (\$3000).
- Jillian Postlethwaite (CU Boulder) 2026 - Present
- Solar system stability due to post-main sequence stellar kicks and mass loss.
 - Co-supervision with Professor Taeho Ryu and Dr. Stephen Majeski.
- Valeria Saldana (CU Boulder) 2025 - Present
- *Project*: Distinguishing bound stellar mass between gravitational wave kick or three-body interactions supermassive black holes.
 - Co-supervision with Professor Taeho Ryu and Dr. Jane Bright.
- Mac Turner (Cornell, computer science undergraduate) 2024 - 2025
- *Project*: Machine learning, loss scaling laws, and noise generation for exoplanet spectra.
 - Co-supervision with Professor Lisa Kaltenegger.

TEACHING EXPERIENCE

COURSE INSTRUCTOR

CU Boulder

- (grad) ASTR 5835: Seminar in Planetary Science — Interstellar Objects 2026

TEACHING ASSISTANTSHIP

University of Toronto

- (undergrad) AST 424: Introduction to Astrophysical Research 2025
- (undergrad) AST 320: Introduction to Astrophysics 2022, 2025
- (undergrad) AST 325, Co2: Practical Astronomy 2022 - 2024
- (undergrad) AST 221: Stars & Planets 2024
- (undergrad, non-major) AST 251: Life on Other Worlds 2022 - 2024
- (undergrad, non-major) AST 201: Introduction to Astronomy 2022
- (undergrad, non-major) AST 101: Introduction to Astronomy 2021

READING PROJECTS

2022, 2023

- Supervised two undergraduate students (2nd & 4th year) on reading projects on cosmology and general relativity through the Physics Directed Reading Program (Toronto).

HIGH SCHOOL STUDENTS

2025

- Supervised Arpit Singh & Tariq Ali-Bowen on a summer project, *Simulating exomoon orbits with the N-body code REBOUND*, funded through [Visions of Science](#).

BLACKBOARD LECTURES

CU Boulder

- Dynamics I: The oldest problem in astrophysics 2026
(Keplerian orbits from Newton's law, Hamiltonian dynamics, symplecticity)
- Dynamics II: Poincaré invariants, action-angle variables, eccentricity vector, the Galactic tide 2026
- Dynamics III: Loss cone dynamics and the Solar-Galactic Hill limit 2026

PUBLIC OUTREACH

SERVICE

CU Boulder Astronomy Day, Volunteer	2026
CU Astronomy Club, Applying to Astronomy Graduate School	2025
Toronto Public Library system, Volunteer	2024
Canadian Young Physicists' Tournament (CaYPT), Juror & Juror Chair	2022 - 2024
Summer Undergraduate Research Program (Toronto), Speaker and Organizer	2024
Toronto Astro On Tap, Volunteer	2022 - 2023
AstroTours, Volunteer	2021 - 2023
Cornell Astronomical Society, Volunteer	2016 - 2020

OUTREACH TALKS

1. [Teaching the Universe](#), Overview on exoplanets (audience: high school teachers) 2025
2. Toronto Public Library, Solar eclipse workshops (five total) 2024
3. Summer Undergraduate Research Program, Overview on exoplanet science 2024
4. Summer Undergraduate Research Program, Grad school admissions overview 2023, 2024
5. Toronto Public Library, The Fate of Comets 2024
6. AstroTours, The Fate of Comets ([YouTube](#)) 2024
7. University of Toronto Family Day Outreach Event, The Oort Cloud 2023

SERVICE

Friday Lunch Seminar (CU Boulder), Organizer	2026 - Present
Journal Reviewer	2025 - Present
• Open Journal of Astrophysics, Astronomy & Astrophysics	
Undergraduate Research Opportunities Program (CU Boulder), Reviewer	2026 - Present
Climate of the Department Committee (Toronto)	2022 - 2024
AstroTours (Toronto), Co-Director	2022 - 2023
Graduate Astronomy Students Association (Toronto)	2020 - 2025
Summer Undergraduate Research Program (SURP , Toronto) Organizer	2023
Local Organizing Committee: SDSS V, CITA Planet Day	2022

MEDIA COVERAGE

- White Dwarfs Eat More Planetary Debris Than Thought, But Magnetic Fields Hide It
[phys.org](#), [Universe Today](#) 2026
- A galactic tide could be pushing comets into collisions with white dwarfs
[BBC Sky at Night](#) 2024
- Mysterious fast radio bursts could be caused by asteroids slamming into dead stars
[space.com](#), [phys.org](#), [NewScientist](#) 2024
- Where's the water?
[Astrobites](#) 2022
- Machine Learning Will be one of the Best Ways to Identify Habitable Exoplanets
[Universe Today](#), [phys.org](#) 2022

PUBLICATIONS

View on the [NASA Astrophysics Data System](#) or [Google Scholar](#).

3 in review, 14 peer-reviewed, 6 first authors, 4 major contributions (second or third author).

IN REVIEW (3)

1. **D. Pham**, S. Majeski et al., 2026, [submitted to MNRAS](#), [arXiv:2609.20941](#).
White Dwarf Natal Kicks as Velocity-Space Random Walks. II. Binary Stellar Evolution and Observational Kick Constraints.
2. S. Majeski, **D. Pham** et al., 2026, [submitted to MNRAS](#), [arXiv:2609.20940](#).
White Dwarf Natal Kicks as Velocity-Space Random Walks. I. Fokker-Planck Theory.
3. B. Vaughan et al. (including **D. Pham**), 2026, [submitted to OJAp](#), [arXiv:2607.01220](#).
TIME Commissioning Observations: II. On-sky Characterization and the 2D Map Data Processing Pipeline.

LEAD AUTHOR (6)

1. **D. Pham**, A. G. Taylor, T. Cunningham, 2026, [ApJ](#), 1008, 257.
The Effects of Magnetic Accretion on the Spatial Extent of White Dwarf Pollution.
2. **D. Pham**, M. Hopkins, et al., 2024, [ApJ](#), 977, 232.
Fast Radio Bursts and Interstellar Objects.
3. **D. Pham**, H. Rein, 2024, [MNRAS](#), 530, 2526.
Polluting White Dwarfs with Oort Cloud Comets.
4. **D. Pham**, H. Rein, D.S. Spiegel, 2024, [OJAp](#), 7, 1.
A new timestep criterion for N-body simulations.
5. **D. Pham**, L. Kaltenegger, 2022, [MNRAS Letters](#), 513, L72.
Follow the water: finding water, snow, and clouds on terrestrial exoplanets with photometry and machine learning.
6. **D. Pham**, L. Kaltenegger, 2021, [MNRAS](#), 504, 6106.

Color Classification of Earth-like Planets with Machine Learning.

SECOND AND THIRD AUTHOR (4)

7. M. Poon, **D. Pham**, M. L. Bryan, et al., 2026, *ApJL*, 1002, L36.
Stellar separation shapes spin-orbit alignment in visual binaries.
8. M. Poon, H. Rein, **D. Pham**, 2024, *OJAp*, 7, 109.
A potential exomoon from the predicted planet obliquity of β Pictoris b.
9. J. Li, L. Kaltenegger, **D. Pham**, D. Ruppert, 2024, *MNRAS Letters*, 527, L137.
Characterization of extrasolar giant planets with machine learning.
10. M. Schlecker, **D. Pham**, R. Burn, et al., 2021, *A&A*, 656, A73.
The Determinism of Global Planet Formation Models.

CONTRIBUTING AUTHOR (4)

11. S. Yang et al. (including **D. Pham**), 2026, *ApJ*, 1003, 23.
TIME Commissioning Observations: I. Mapping Dust and Molecular Gas in the Sgr A Molecular Cloud Complex at the Galactic Center.
12. M. Poon et al. (including **D. Pham**), 2025, *ApJL*, 994, L48.
Early Evidence for Isotropic Planetary Obliquities in Young Super-Jupiter Systems.
13. V. Butler et al. (including **D. Pham**), 2024, *SPIE Proceedings*, 13102, 156.
TIME, the Tomographic Ionized-carbon Mapping Experiment: an update on design, characterization, and data from the 2022 commissioning observations.
14. A. Harding et al. (including **D. Pham**), 2018, *MNRAS*, 475, 79.
Predicting Gravitational Lensing by Stellar Remnants.